🏆 Infosys SE Elite Crack Vault — 09 Mensuration 2D 3D and Coordinate Geometry Elite

> **Target Role: Infosys Systems Engineer (SE)** (3.6 – 4.0 LPA Recruitment Drive)
> **Curation Standard: 100% Pure SE Role Specific** (All SP/DSE Upper-Level & Practice Variants Purged)
> **Total Curated Questions in Topic: 10 High-Yield Questions**
> **Question Breakdown:** 🔴 `5 Must-Solve` | 🟡 `5 Core SE Level` | ⚪ `0 Practice Variant`
> **Verification Score:** 100% Mathematically & Deductively Proven with Zero Hallucination

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Problem 01: Cube Root & Mensuration Shortcuts 🔴 `[🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]`

> 🎯 **Exam Priority:** 🔴 `[🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]`
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)
> **Question ID:** `SEC02_BOOK_P05_Q30` | **Source File:** `250 + Video Solution Quant Assorted Book.pdf`

Question Statement:

Find the cube root of 328,509:

$$\sqrt[3]{328509}$$

Options

- a. 69
- b. 50
- c. 97
- d. 78

Correct Answer

> [!IMPORTANT] **Verified Answer Key**: `A. 69`

Step-by-Step Mathematical Proof & Arithmetic Solution

1. **Units Digit Determination:** The number ends in 9. The only single-digit number whose cube ends in 9 is 9 ($9^3 = 729$).

$$\text{Units digit of } \sqrt[3]{328509} = 9$$

2. **Tens Digit Determination:** Strike out the last three digits (509), leaving the prefix number 328. Determine the largest integer n such that $n^3 \leq 328$:

$$6^3 = 216 \leq 328 < 7^3 = 343$$

   Therefore, the tens digit is **6**.

3. **Verification:**

$$69^3 = 69 \times 69 \times 69 = 4,761 \times 69 = 328,509$$

   Therefore, the cube root is **69**. Verified Answer is **A**.

Problem 02: Cube Root & Mensuration Shortcuts 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)
> **Question ID:** `SEC02_BOOK_P05_Q31` | **Source File:** `250 + Video Solution Quant Assorted Book.pdf`

Question Statement:

Find the cube root of 50,653:

$$\sqrt[3]{50653}$$

Options

- a. 22
- b. 37
- c. 97
- d. 69

Correct Answer

> [!IMPORTANT] **Verified Answer Key**: `B. 37`

Step-by-Step Mathematical Proof & Arithmetic Solution

1. **Units Digit Determination:** The number ends in 3. The only single-digit number whose cube ends in 3 is 7 ($7^3 = 343$).

$$\text{Units digit of } \sqrt[3]{50653} = 7$$

2. **Tens Digit Determination:** Strike out the last three digits (653), leaving the prefix number 50. Determine the largest integer n such that $n^3 \leq 50$:

$$3^3 = 27 \leq 50 < 4^3 = 64$$

   Therefore, the tens digit is **3**.

3. **Verification:**

$$37^3 = 37 \times 37 \times 37 = 1,369 \times 37 = 50,653$$

   Therefore, the cube root is **37**. Verified Answer is **B**.

Problem 03: Angular Geometry & Directional Rotation 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)
> **Question ID:** `SEC02_BOOK_P26_Q199` | **Source File:** `250 + Video Solution Quant Assorted Book.pdf`

Question Statement:

Shiva is facing North. He turns 180° and 45° in the clockwise direction, then 270° in the anti-clockwise direction, and finally turns 180°. What is his facing direction now?

Options

- a. SE (South-East)
- b. SW (South-West)
- c. South
- d. NE (North-East)

Correct Answer

> [!IMPORTANT] **Verified Answer Key**: `A. SE (South-East)`

Step-by-Step Mathematical Proof & Arithmetic Solution

1. **Angular Reference Frame:** Let North be 0° (or 360°). Clockwise (CW) rotations are positive ($+$) and Anti-Clockwise (ACW) rotations are negative ($-$).
2. **Net Angular Calculation:**

$$\text{Initial Position} = 0^\circ \ (\text{North})$$

$$\Delta\theta_1 = +180^\circ \ (\text{CW})$$

$$\Delta\theta_2 = +45^\circ \ (\text{CW})$$

$$\Delta\theta_3 = -270^\circ \ (\text{ACW})$$

$$\Delta\theta_4 = +180^\circ \ (\text{CW or general reverse})$$

3. **Sum of Angles:**

$$\theta_{\text{final}} = 0^\circ + 180^\circ + 45^\circ - 270^\circ + 180^\circ = 135^\circ$$

4. **Compass Mapping:**
   - 0° = North
   - 90° = East
   - 135° = South-East (SE)
   - 180° = South Therefore, Shiva is facing **South-East (SE)**. Verified Answer is **A**.

Problem 04: 2D Perimeter & Wire Re-bending Invariant 🔴 `[🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]`

> 🎯 **Exam Priority:** 🔴 `[🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]`
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)
> **Question ID:** `SEC02_MENSUR_WIRE_Q01` | **Source File:** `CampusMonk Quantitative Aptitude`

Question Statement:

A copper wire is bent in the form of a circle of radius 28 cm. If the same wire is straightened and re-bent into the shape of a square, what is the length of each side of the square? (Take $\pi = 22/7$)

Options

- a. 40 cm
- b. 44 cm

- c. $48\ \text{cm}$
- d. $52\ \text{cm}$

Correct Answer

> [!IMPORTANT] **Verified Answer Key**: `B. 44 cm`

Step-by-Step Mathematical Proof & Arithmetic Solution

1. **Length Conservation Invariant:** When a wire of fixed length is re-bent from one shape to another, the perimeter remains strictly invariant:

$$\text{Perimeter of Square} = \text{Circumference of Circle}$$

2. **Circumference of the Circle:**

$$C = 2\pi r = 2 \times \frac{22}{7} \times 28 = 2 \times 22 \times 4 = 176\ \text{cm}$$

3. **Side of the Square (a):**

$$4a = 176 \implies a = \frac{176}{4} = 44\ \text{cm}$$

Therefore, the length of each side of the square is **44 cm**. Verified Answer is **B**.

Problem 05: 3D Cuboid Surface Area & Ratio Scaling 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]

> 🎯 **Exam Priority:** 🔴 [🔥 MUST-SOLVE | INFOSYS SE ACTUAL PATTERN]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)
> **Question ID:** `SEC02_TEST_MENSUR_Q01` | **Source File:** `Mensuration Pre-Test_QUESTIONS.json`

Question Statement:

The ratio of the length, width, and height of a closed rectangular cuboid is $7 : 5 : 4$. If its entire surface area (Total Surface Area) is $2,656\ \text{cm}^2$, find its width.

Options

- a. $16\ \text{cm}$
- b. $20\ \text{cm}$
- c. $24\ \text{cm}$
- d. $28\ \text{cm}$

Correct Answer

> [!IMPORTANT] **Verified Answer Key**: `B. 20 cm`

Step-by-Step Mathematical Proof & Arithmetic Solution

1. **Dimension Modeling:** Let the dimensions of the cuboid be:

$$\text{Length } l = 7x, \quad \text{Width } w = 5x, \quad \text{Height } h = 4x$$

2. **Total Surface Area (TSA) Formula:**

$$\text{TSA} = 2(lw + wh + hl)$$

$$\text{TSA} = 2[(7x \times 5x) + (5x \times 4x) + (4x \times 7x)]$$

$$\text{TSA} = 2[35x^2 + 20x^2 + 28x^2] = 2[83x^2] = 166x^2$$

3. **Equating to Given Area:**

$$166x^2 = 2656$$

$$x^2 = \frac{2656}{166} = 16 \implies x = \sqrt{16} = 4$$

4. **Calculating Width (w):**

$$w = 5x = 5 \times 4 = 20 \text{ cm}$$

(Also: length $= 7 \times 4 = 28$ cm, height $= 4 \times 4 = 16$ cm). Therefore, the width is **20 cm**. Verified Answer is **B**.

Problem 06: Cylinder Total Surface Area 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)
> **Question ID:** SEC02_TEST_MENSUR_Q02 | **Source File:** Mensuration Pre-Test_QUESTIONS.json

Question Statement:

Find the total surface area (in cm^2) of a right circular cylinder having a diameter of 7 cm and a height of 6 cm. (Take $\pi = 22/7$)

Options

- a. 185 cm^2
- b. 198 cm^2
- c. 209 cm^2
- d. 220 cm^2

Correct Answer

> [!IMPORTANT] **Verified Answer Key**: C. 209 cm²

Step-by-Step Mathematical Proof & Arithmetic Solution

1. **Given Parameters:**
   - Diameter $d = 7$ cm $\implies$ Radius $r = \frac{7}{2} = 3.5$ cm
   - Height $h = 6$ cm
2. **Formula for Total Surface Area of Cylinder:**

$$\text{TSA} = 2\pi r(r + h)$$

3. **Arithmetic Substitution:**

$$\text{TSA} = 2 \times \frac{22}{7} \times \frac{7}{2} \times \left(\frac{7}{2} + 6\right)$$

$$\text{TSA} = 22 \times (3.5 + 6) = 22 \times 9.5 = 209 \text{ cm}^2$$

Therefore, the total surface area is **209 cm²**. Verified Answer is **C**.

Problem 07: 3D Sphere Melting & Recasting Conservation 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)
> **Question ID:** `SEC02_MENSUR_RECAST_Q01` | **Source File:** `CampusMonk Quantitative Aptitude`

Question Statement:

A solid metallic sphere of radius 6 cm is melted and recast into small spherical lead balls, each of radius 2 cm. How many such small balls can be made from the parent sphere?

Options

- a. 18
- b. 24
- c. 27
- d. 36

Correct Answer

> [!IMPORTANT] **Verified Answer Key**: `C. 27`

Step-by-Step Mathematical Proof & Arithmetic Solution

1. **Volume Conservation Invariant:** When a metal body is melted and recast, total volume is conserved:

$$V_{\text{parent}} = n \times V_{\text{small}}$$

2. **Volume of a Sphere Formula:**

$$V = \frac{4}{3}\pi R^3$$

3. **Ratio of Volumes:**

$$n = \frac{\frac{4}{3}\pi R^3}{\frac{4}{3}\pi r^3} = \left(\frac{R}{r}\right)^3$$

4. **Substitution:**

$$n = \left(\frac{6}{2}\right)^3 = 3^3 = 27$$

Therefore, exactly **27** small spherical balls can be manufactured. Verified Answer is **C**.

Problem 08: Cone and Cylinder Equal Base Volume Comparison 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)
> **Question ID:** SEC02_MENSUR_CONE_CYL_Q01 | **Source File:** CampusMonk Quantitative Aptitude

Question Statement:

A right circular cone and a cylinder have equal circular bases and identical heights. If the volume of the cylinder is $180\ \text{cm}^3$, what is the volume of the cone?

Options

- a. $45\ \text{cm}^3$
- b. $60\ \text{cm}^3$
- c. $90\ \text{cm}^3$
- d. $120\ \text{cm}^3$

Correct Answer

> [!IMPORTANT] **Verified Answer Key**: B. 60 cm³

Step-by-Step Mathematical Proof & Arithmetic Solution

1. **Geometric Relationship:** For identical base radius r and identical height h:

$$V_{\text{cylinder}} = \pi r^2 h$$

$$V_{\text{cone}} = \frac{1}{3}\pi r^2 h = \frac{1}{3}V_{\text{cylinder}}$$

2. **Evaluation:**

$$V_{\text{cone}} = \frac{1}{3} \times 180 = 60\ \text{cm}^3$$

Therefore, the volume of the cone is **60 cm³**. Verified Answer is **B**.

Problem 09: Circular Track Concentric Pathway Area 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]

> 🎯 **Exam Priority:** 🟡 [⭐ CORE SE LEVEL | HIGH PROBABILITY]
> 📊 **Target Difficulty:** Infosys SE Standard Level (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)
> **Question ID:** SEC02_MENSUR_TRACK_Q01 | **Source File:** CampusMonk Quantitative Aptitude

Question Statement:

A circular garden of radius $21\ \text{m}$ is surrounded by a running track of uniform width $3.5\ \text{m}$. What is the area of the running track? (Take $\pi = 22/7$)

Options

- a. $462.5\ \text{m}^2$
- b. $485.0\ \text{m}^2$

- c. $500.5\ \text{m}^2$
- d. $525.0\ \text{m}^2$

Correct Answer

> [!IMPORTANT] **Verified Answer Key**: `C. 500.5 m²`

Step-by-Step Mathematical Proof & Arithmetic Solution

1. **Concentric Radii:**
   - Inner radius $r = 21\ \text{m}$
   - Outer radius $R = r + \text{width} = 21 + 3.5 = 24.5\ \text{m}$
2. **Annulus (Track) Area Formula:**

$$\text{Area} = \pi(R^2 - r^2) = \pi(R - r)(R + r)$$

3. **Calculation:**

$$\text{Area} = \frac{22}{7} \times (24.5 - 21) \times (24.5 + 21)$$

$$\text{Area} = \frac{22}{7} \times 3.5 \times 45.5$$

Since $\frac{3.5}{7} = 0.5$:

$$\text{Area} = 22 \times 0.5 \times 45.5 = 11 \times 45.5 = 500.5\ \text{m}^2$$

Therefore, the area of the track is **500.5 m²**. Verified Answer is **C**.

Problem 10: Trapezium Field Geometry 🟡 [⭐ `CORE SE LEVEL | HIGH PROBABILITY`]

> 🎯 **Exam Priority:** 🟡 [⭐ `CORE SE LEVEL | HIGH PROBABILITY`]
> 📊 **Target Difficulty:** `Infosys SE Standard Level` (Single/Double Step Deduction)
> 💡 **Preparation Advice: MANDATORY FOR 5-6 NOV DRIVE** (Direct recurring exam concept)
> **Question ID:** `SEC02_MENSUR_TRAP_Q01` | **Source File:** `CampusMonk Quantitative Aptitude`

Question Statement:

The parallel sides of an isosceles trapezium are $20\ \text{cm}$ and $10\ \text{cm}$. Each of the non-parallel oblique sides is $13\ \text{cm}$. What is the total area of the trapezium?

Options

- a. $150\ \text{cm}^2$
- b. $180\ \text{cm}^2$
- c. $210\ \text{cm}^2$
- d. $240\ \text{cm}^2$

Correct Answer

> [!IMPORTANT] **Verified Answer Key**: `B. 180 cm²`

Step-by-Step Mathematical Proof & Arithmetic Solution

1. **Determining Height of Trapezium (h):** In an isosceles trapezium with parallel sides $a = 20\ \text{cm}$ and $b = 10\ \text{cm}$, dropping perpendiculars from the top corners to the base leaves two symmetric right triangles with base:

$$x = \frac{a - b}{2} = \frac{20 - 10}{2} = 5\ \text{cm}$$

2. **Applying Pythagorean Theorem for Height (h):** The hypotenuse is the non-parallel side $= 13\ \text{cm}$.

$$h = \sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = 12\ \text{cm}$$

3. **Trapezium Area Formula:**

$$\text{Area} = \frac{1}{2} \times (a + b) \times h$$

$$\text{Area} = \frac{1}{2} \times (20 + 10) \times 12 = 15 \times 12 = 180\ \text{cm}^2$$

Therefore, the area of the trapezium is **180 cm²**. Verified Answer is **B**.
